

Privileged Brain fine tuning

A small local code model specialized for system operations

The selected model is run7_cot_q4km.gguf, based on Qwen2.5-Coder-1.5B-Instruct. LoRA supervised fine-tuning adapted the model to command generation, refusal and clarification. The work targeted the Privileged Brain; the production Quarantined Brain configuration uses Gemini.

Training objective and data

Training examples map natural-language administration requests to a two-line answer: a short visible rationale followed by COMMAND. The command field contains a shell command, REFUSE or CLARIFY. These supervised rationales are output text; they are not proof of an internal reasoning process.

The checked-in structured-rationale corpus contains 15,377 records: 13,839 training and 1,538 validation examples. The processor filters malformed records and known dangerous commands in non-refusal targets, then shuffles and splits the data. The current files provide dataset evidence; they are not a complete historical run manifest.

Checked-in SFT configuration

Setting	Value
Base model	Qwen/Qwen2.5-Coder-1.5B-Instruct; bfloat16 loading
LoRA adapters	Rank 8; alpha 16; dropout 0.05; attention and MLP projections
Optimization	3 epochs; AdamW; learning rate 2e-4; cosine schedule
Batch and context	Micro-batch 2 × gradient accumulation 8 = effective batch 16; maximum sequence length 512
Checkpoint selection	Evaluate and save every 500 steps; retain the checkpoint with lowest validation loss

Why the target format mattered

Earlier runs produced explanatory prose or multi-command answers and showed incomplete refusal coverage. Run 7 used a structured rationale and a distinct command field. This changed the supervised representation and made the output easier to parse. The historical record reports improved refusal results; it does not isolate causality through a controlled ablation.

From adapter to local inference

The adapter was merged into the base model, converted to GGUF and quantized to Q4_K_M. The reported artifact size is 940 MB, with SHA-256 recorded in the model manifest. Deployment uses llama-server; the Controller prompts for structured MCP tool calls and validates them. The training corpus described here is NL-to-Bash, not a demonstrated native MCP training corpus.

Sources: privileged-brain/v2/scripts/{sft_train.py,process_datasets.py,fuse_lora.py}; privileged-brain/v2/data/processed/cot/docs/implementation_plan.md, Phase 4 close-out; models/checksums.sha256.

Measured results

Run 7 was retained after a later candidate regressed on safety

The June 2026 Phase 4 close-out records the results below for the Run 7 Q4_K_M artifact. The tests measure separate properties: refusal behavior, output structure, lexical agreement and fixture-based execution equivalence. No one metric establishes overall task accuracy.

Recorded Run 7 evaluation

Measure	Result	Test scope
Adversarial refusal	100% · 20/20	Fixed dangerous-prompt suite; minimum release threshold 19/20.
MCP JSON validity	95.5%	132 generations across 22 tool names; structural output test.
Exact command match	7.7%	Normalized string equality on 300 held-out examples.
Command token F1	53.6%	Mean lexical overlap with reference commands on the held-out set.
Execution equivalence	23.3%	Successful return codes, stdout and file state agree on the runnable subset.

The Run 8 release decision

Run 8 continued training on 1,027 curated examples plus approximately 3,000 replay examples, at learning rate $5e-5$ for two epochs. Exact match improved to 12% and token F1 to 55.9%. Adversarial refusal fell to 90% (18/20), including unsafe permission-change outputs for protected system paths.

Decision: reject Run 8 and retain Run 7. The loss of refusal behavior outweighed the modest utility gains. DPO preference training exists in the pipeline and was used in earlier experiments; the selected Run 7 release was SFT-only.

What the results establish

The 20/20 result shows complete refusal on that fixed suite, not general safety across unseen requests. The MCP evaluator checks parseability, required fields and catalogue membership; its tool check does not establish exact agreement with the allowed tool or full parameter correctness.

Valid alternative shell commands can reduce exact-match scores, but the low utility figures still require broader validation. Raw Run 7 result JSON is absent from the reviewed tree; the reported values are supported by the project close-out and checked-in evaluation code.

Sources: docs/implementation_plan.md, Phase 4 close-out and Run 8 comparison; privileged-brain/v2/scripts/{phase4_eval.py,eval_adversarial_gguf.py,dpo_train.py}. Metrics are historical project records, not new measurements for this fact sheet.